

MaRS Recruitment 2026

Task #1 Science Report

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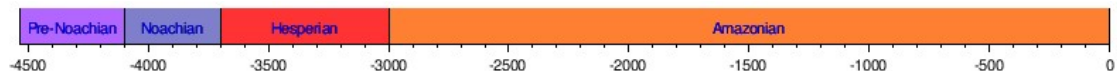
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[GitHub Repository Link](#)

1. Crater Density Timescale

The geological time period of Mars is divided into 4 main timelines. The crater impact sites will be classified into 3 timelines instead of 4 for the sake of simplicity:

- **Pre – Noachian Period:** The interval between the accretion (formation) and the differentiation (distribution of minerals) of the planets about **~4.5 billion years ago (Gya)** to the formation of the **Hellas Basin**, between **4.1 to 3.8 Gya**. The geological record of this time period has been erased due to natural erosion and impacts due to meteors. This period will not be included in the classification of crater timescales. This is during the time of the **Late Heavy Bombardment** era.
- **Noachian Period:** Time period of the formation of the oldest surfaces of Mars around **4.1 to 3.7 Gya**. Craters of this time period are large which have scarred the surface. Liquid water in the form of large lakes or oceans is theorized to exist which can be reason for the erosion causing the **Tharsis Bulge**.
- **Hesperian Period: Approximately 3.7 – 3.0 Gya.** Marked by extensive plains of lava as the formation of **Olympus Mons** began during this period. Violent flow of water carved out channels around the **Chryse Planitia** (Located in the Northern lowlands).
- **Amazonian Period: 3.0 Gya to present.** This era of regions has significantly fewer impact craters varied. Flow of lava, glaciers and liquid water have existed during this period.



This time period will focus on dividing the crater impact sites into 3 main timelines which are the **Noachian**, **Hesperian** and **Amazonian** periods.

Period 1 – Noachian (4.1-3.7 Gya)

This was part of the period during the Late Heavy Bombardment with numerous asteroid impacts creating the largest craters on Mars till date.

Crater	Diameter	Reasoning
Hellas Planitia Crater	2300km	Largest visible impact basin, rim is heavily degraded. The extreme depth indicates formation during peak bombardment. Ejecta rays are kilometers thick but covered due to volcanic activity and erosion.
Argyre Planitia Crater	1800km	Walls are heavily eroded and overlapped by multiple smaller craters indicating an older age. The multi-ring impact structure formed during early Noachian era with a smaller crater buried inside it.

		Original ejecta patterns are no longer identifiable, reworked over ecological time.
Huygens Crater	460km	Ring structure is well preserved. The impact exposed pre-Noachian minerals as deep as 25km below the surface revealing ancient material buried within the surface penetrating thick layers of Hellas ejecta. Crater served as a basin for aqueous activity including valley networks and surface alteration now exposed by smaller impacts.
Schiaparelli Crater	459km	Contains layered deposits formed by dusty winds, volcanoes or water flows, thereby indicating ancient age with aqueous deposition. Original ejecta are nowhere to be found. Rover data indicate layered deposits in the interior formed by wind, volcanic lava or water deposits. Layered patterns suggest climate cycles driven by variations on Mars' tilt.
Cassini Crater	415km	Heavily filled floor with rim topography indicating prolonged erosion of soil. No signs of ejecta blankets. Evidence of open-basin lakes discovered, consistent with Noachian wet climatic period.

Period 2 – Hesperian (3.7-3.0 Gya)

The Hesperian era is characterized by volcanic activity and massive floods thereby carving outflow channels transitioning from a water world to the dry, dusty and cold planet presently.

Crater	Diameter	Reasoning
Gusev Crater	166km	Partially filled rim indicates post-impact volcanic flooding. Presence of lava indicates Hesperian signature at the crater floor confirmed by Spirit rover. Original ejecta have been buried under lava flows. Spirit rover confirmed the cratered plains of Gusev are
Jezero Crater	49km	Delta deposits date to ~3.5 – 3.9 Gya. River channels have cut into the rim which indicate Hesperian era activity. Perseverance rover confirmed delta deposits within the crater.
Holden Crater	141km	Smaller craters superposed on rim but not as densely as Noachian craters. Crater is partially filled with alluvial fan deposits indicating water activity during the Hesperian era. Ejecta are partially preserved by modified significantly due to catastrophic flooding.
Miyamoto Crater	150km	Heavily infilled floor with subdued topography. Opportunity rover displayed sulphate minerals

		along with acidic water signatures in the deposits. Original ejecta blankets are modified by aqueous processes.
Colombus Crater	100km	Rim is heavily degraded, less than Noachian craters but more than Amazonian ones. Curiosity rover revealed sulphate and phyllosilicate mineral layers in the crater. Findings are consistent with Hesperian episodic water activity.

Period 3 – Amazonian (3.0 Gya - Present)

The Amazonian era is characterised by sparse crater density, volcanic activity ceased during this time period. Crater impact's surface features are preserved due to low impact rates during this period.

Crater	Diameter	Reasoning
Zunil crater	10km	Sharp rim which is well preserved. Created almost 10 million secondary craters (diameter around 10-200m each) extending upto 1600km. Craters show bright ejecta indicating less modification since formation.
Mojave Crater	58km	Fresh morphology with sharp rims and no observable degradation. Radial patterns of ejecta visible with thermal infrared rays. Zero superposed craters on ejecta blankets confirming geologically recent formation.
Corinto Crater	14km	No measurable erosion with pristine morphology. Distinct thermal rays covering 2000km with secondary craters that are fresh and unmodified by aeolian activity due to young ages with characteristic bright rounded ejecta blankets.
Tomini Crater	2km	Small, sharp rim with no erosion. Located on Amazonian terrain. Ejecta blankets are preserved from orbit. No secondary impacts on the ejecta signifying recent impact.
Zumba Crater	3.5km	Fresh sharp rims with fresh small craters on Mars. Ejecta deposits are highly detailed with single layered ejecta deposits indicating volatile rich subsurface interaction at the time of impact.

2. Landing Site selection : Mission “Red Genesis”

Before choosing a site we need to verify two parameters:

- Elevation must be below 1000m: For atmospheric density to deploy the parachute in order to bring the aircraft to correct terminal velocity and provide enough density and airflow to bring the aircraft to the correct terminal velocity.
- Latitude must be between 30 degrees N and 30 degrees South to receive solar power to power the rover.

My proposed site of landing is the **Ancient River Delta within Isidis Planitia**. The elevation of the location around 3500m below the mean surface level mark of Mars. The latitude coordinates of the location are 18 degrees Northward, thereby fulfilling our constraints.

River deltas are excellent for seeking the evidences of life that have been deposited and preserved for millions of years – ancient organic molecules and signs of microbial life can be collected on lakebeds and delta sediments.

Deltas are good at preserving biochemical signatures which means that evidence of life could have existed in the water, at the interface between the sediments and water, or things swept in from rivers and deposited in the delta fan.

Ancient river delta systems offer promising rock types simultaneously. Scientists are interested in carbonates which help preserve fossilized signs of ancient life, alongside clay minerals which are formed only in the presence of liquid water.

But this brings multiple hazards as well. Ancient river delta terrain are covered with landing hazards such as cliffs overlooking delta scapes, dune fields with quicksand and large rocks requiring precise landing to avoid these obstacles.

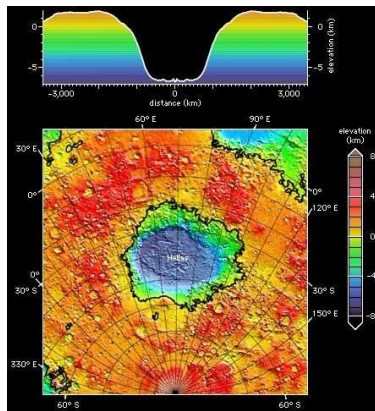
3. Hellas Planitia

The Origin of Hellas Planitia

Hellas Planitia is not a volcano like Olympus Mons rather a massive impact basin that was formed by a catastrophic asteroid collision during the early periods of Mars, likely during the Late Heavy Bombardment era.

When the asteroid struck Mars, an enormous amount of kinetic energy was released. This impact excavated deep layers of crust thereby ejecting huge amounts of rock and debris outward hence creating a ~300km wide crater that is almost 7km deep.

Volcanic Calderas are formed separately. Calderas are created when magma beneath a volcano empty which cause the surface above the magma chamber to collapse inward. Hellas Planitia shows characteristics of an impact structure rather than volcanic collapse, such as ejecta deposits, shock related geological deformation and large scale crust expulsion.



[Image source](#)

Regional Dust Storms

Hellas Planitia experiences intense dust storms which are frequent in the area. Due to the extreme depth, the pressure inside the basin is higher compared to the surrounding Martian terrain. This created a strong atmosphere that is capable of lifting and trapping dust for extended periods of time.

These storms reduce visibility and solar irradiance, thereby making rover navigation and power generation through solar extremely difficult.

Subsurface Ice and Frost Deposits

Scientists have also identified evidence of buried glaciers and frost deposits within Hellas Planitia. The basin's extremely low elevation allows cold and dense air to gather which help to preserve water beneath the rocks.

These ice deposits may provide important details and clues about the planet's climatic history and the frost deposits within the basin and will also help to determine the long term stability and life of water on the planet.

Future Rover Mission

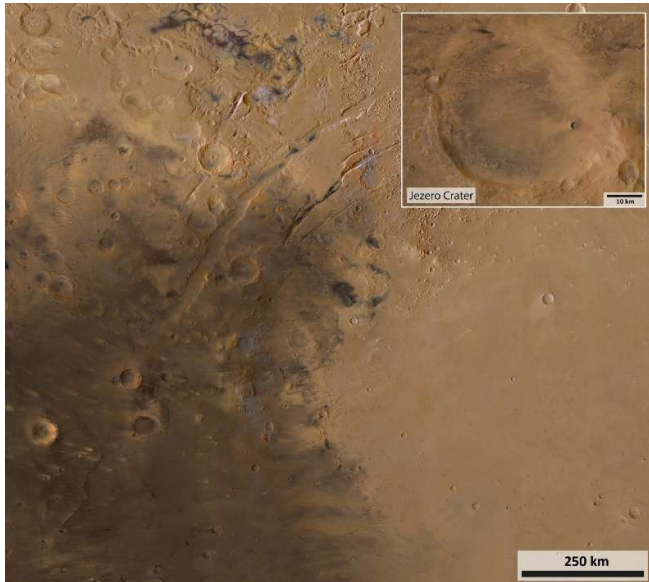
A future rover mission that explores Hellas Planitia should carry a Ground Penetrating Radar system. This piece of equipment will allow scientists to investigate hidden subsurface layers beneath the Martian regolith and thereby help to detect buried glaciers, underground ice deposits, sedimentary structures and ancient geological formations that are hidden below the basin floor.

Studying these underground features may help scientists better understand the evolution of the Martian climate and determine whether water remained stable in the region for long periods of time.

4. Data Based Observation

Landing Site

For mission Argo – 1, I select the Jezero Crater as my preferred site of landing.



[Image Link](#)

Scientific Justification

Jezero Crater is one of the most scientifically valuable locations on Mars as it once was a lakebed containing a river delta system. This crater contains sedimentary deposits that may preserve ancient biosignatures and chemical evidence of microbial life.

The presence of clay minerals and carbonates indicates that water existed in the region for a long time. Since water is essential for life as we know, the minerals found in Jezero Crater make it an excellent location for studying the habitability of early Mars.

Comparison with Hellas Planitia

Comparing Jezero Crater with Hellas Planitia – it also faces relatively less extreme environmental conditions. The temperature range is more manageable than Hellas Planitia and the terrain is also far safer for rover operations.

Hellas Planitia remains significantly important as it consists of large ice deposits and experiences higher atmospheric pressure due to lower elevation. These characteristics provide insights to the climatic evolution of Mars.

However, Hellas Planitia also presents several engineering challenges. The basin experiences severe dust storms and extremely low temperatures as well as difficult terrain navigation. These factors are hazardous for the operation of the rover systems as well as power generation.

The crater offers a better balance between scientific value as well as engineering safety.

Risks and Challenges

Despite its advantages, the Jezero Crater still possesses hazards such as rocky and difficult terrain, loose regolith and excessive dust which can damage rover wheels and clog sensitive sensors as well as reduce solar panel efficiency.

The scientific benefits of investigating an ancient lakebed outweighs these risks. The possibility of discovering preserved evidence of microbial life makes Jezero Crater the stronger choice for Mission Argo – 1.

5. Mechanical Modelling

This question involved practical 3D modelling of engineering components in Autodesk Fusion 360. Five models are successfully completed, I ran across difficulties using the loft function for the 5th model. All models have been made with design history enabled throughout the process. The Fusion 360 files (in f3d, DWG and STL formats) are available in my [GitHub Repository](#).

6. Hellas Planitia Anomaly Survival and Prototyping

This question involved practical electronics prototyping on Tinker CAD circuits. All the three modules were successfully built and simulated with C++ code written for each module. The engineering briefs, circuit diagrams and schematics along with public Tinker CAD links are documented in GitHub and uploaded to the project [GitHub Repository](#).

7. BONUS: Four Landing Sites within Hellas Planitia

Introduction:

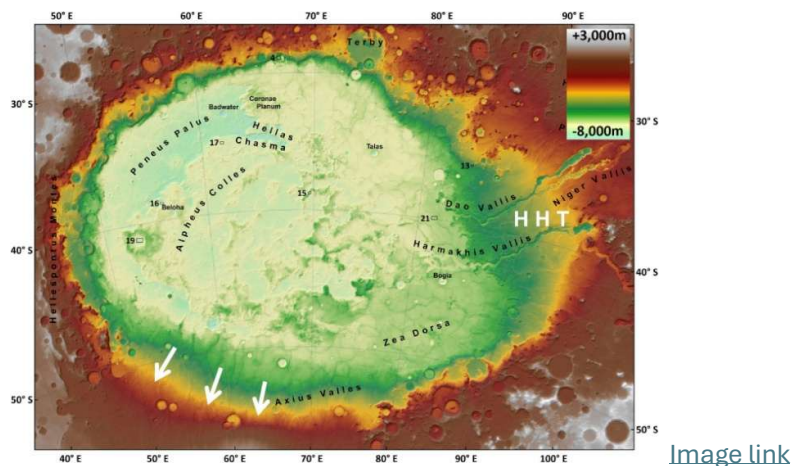
Hellas Planitia, the largest visible impact basin on Mars spans approximately 2300 km across in diameter and reaches depths exceeding 7km below the Martian datum. Its extreme depth can create unique atmospheric and geological conditions that make it scientifically compelling for future exploration. Four strategic landing zones within the basins are proposed below.

Site 1 – Northwestern Basin Floor (Subsurface Ice investigation)

Coordinates: -40 ° S, 60 ° E

Elevation: Approx 7000m

The Northwestern basin represents the deepest accessible terrain within the Hellas region. The atmospheric pressure reaches 1155 Pa which is the highest pressure on Martian surface. This elevated pressure increases the stability range for liquid water to be existing in this region most likely. Subsurface radar data from SHARAD and MARSIS instruments aboard the Mars Reconnaissance Orbiter and Mars Express have identified the potential rich deposits of ice beneath the regolith in this region. A landed mission here carrying a ground penetrating radar and drill system could directly characterize the depth, extent and composition of these deposits thereby addressing the fundamental questions about Mars' water inventory and potential subsurface inhabitability.



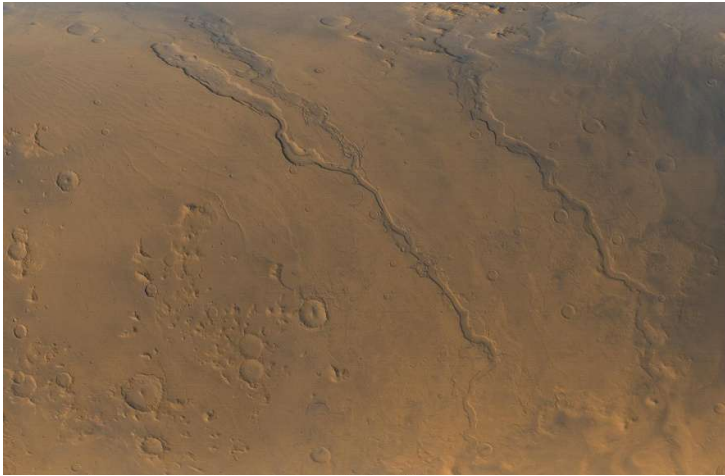
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Site 2 – Hellas Eastern Rim – Dao Vallis Outflow channel

Coordinates: -38 ° S, 92 ° E

Elevation: -4500m

Dao Vallis is one of the several large outflow channels that drain into the eastern margin of Hellas Planitia. These channels were carved by catastrophic fluid releases of either water or Carbon Dioxide during the Hesperian period. The channel mouth where it dives into the basin consists of layered sedimentary deposits that represent multiple flooding events. These sediments are scientifically valuable as they preserve records of Hellas' hydrological history. A rover equipped with a drill and a spectrometer can sample each layer individually and reconstruct the timing of the cataclysmic flooding events. Additionally, the channel walls expose deep crystals that are rarely accessible anywhere else in Mars.

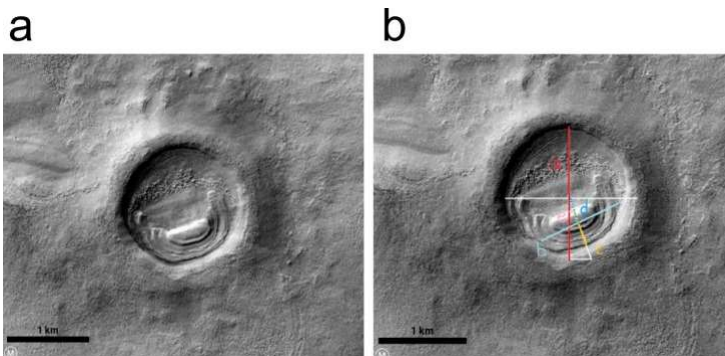
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Site 3 – Southern Basin – Possible Ice Cemented terrain

Coordinates: -49 ° S, 70 ° E

Elevation: -6500m

The southern interior of Hellas shows morphological evidence of glacial and periglacial processes including lobate debris aprons, concentric crater fill, and terrain softening — all indicators of ice-rich subsurface material. This region experiences the coldest temperatures within the basin, reaching -130°C during winter, causing CO_2 and water ice to precipitate seasonally. A lander here carrying a thermal evolved gas analyser could characterise the volatile content of the regolith, while a seismometer would exploit the basin's unique crustal structure to probe interior layering. This site directly addresses questions about Mars' recent glacial history and polar volatile cycles.

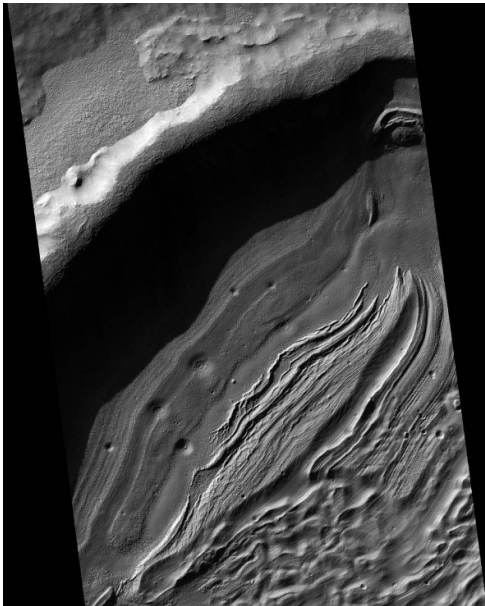
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Site 4 – Northwestern Rim Interior – Impact Melt Rocks

Coordinates: -35 ° S, 55 ° E

Elevation: -3000m

The inner rim scarps of Hellas preserve some of the oldest exposed crustal material on Mars, uplifted and excavated during the original basin-forming impact approximately 4.1 billion years ago. Impact melt rocks are exposed along the rim walls which represent the samples of the early Martian crust that is unavailable anywhere else at the surface. A rover equipped with a spectrometer and X-ray diffractometer could classify the minerals of these ancient materials, configuring the bulk composition of the early Martian crust and the conditions of the impact event itself. This site also offers elevated terrain within the basin providing a partial protection from dust storm deposition while maintaining access to the basin interior.



[Image Link](#)

Disclaimer

AI assistance was used in a limited and specific capacity during the preparation of this report. Specifically, AI tools were used to suggest an initial list of crater names for research purposes, to recommend relevant online resources and references, and to format the Bibliography and Glossary sections. All research and written content presented in this report is entirely my own work. All practical tasks including CAD modelling in Autodesk Fusion 360 and electronics prototyping in Tinkercad were independently completed by me. AI-generated content was verified against primary sources before use.

Glossary

- **Crater Density** — number of impact craters per unit area used to estimate surface age
- **Ejecta** — material thrown out during an impact event
- **Noachian** — earliest Martian geological period (~4.1–3.7 Gya)
- **Hesperian** — middle Martian geological period (~3.7–3.0 Gya)
- **Amazonian** — current Martian geological period (3.0 Gya–present)
- **Regolith** — loose surface material covering bedrock
- **Phyllosilicates** — clay minerals formed in presence of liquid water
- **LiDAR** — Light Detection and Ranging sensor
- **EDL** — Entry Descent and Landing
- **E-box** — Electronics Box housing rover's critical systems
- **Sedimentary Deposits** — layers of material such as sand, silt, clay, and minerals that have been transported by water or wind and deposited over time, potentially preserving chemical and biological signatures of past environments
- **SHARAD** — Shallow Radar, a ground-penetrating radar instrument aboard NASA's Mars Reconnaissance Orbiter designed to detect subsurface ice and rock layers up to 1 km deep
- **MARSIS** — Mars Advanced Radar for Subsurface and Ionosphere Sounding, a radar instrument aboard ESA's Mars Express orbiter capable of probing subsurface structure to depths of several kilometres
- **Late Heavy Bombardment** — a period approximately 4.1–3.8 billion years ago during which the inner solar system experienced an intense spike in asteroid and comet impacts, responsible for forming many of Mars' largest ancient basins
- **River Delta Sediments** — fine-grained material deposited at the mouth of a river where it enters a standing body of water, forming fan-shaped deposits that are considered prime locations for preserving biosignatures
- **Lakebeds** — flat sedimentary deposits formed at the bottom of ancient lakes, preserving a record of past aqueous chemistry and potential microbial activity
- **Ejecta Rays** — bright elongated streaks of material radiating outward from a fresh impact crater, visible from orbit and indicating geologically recent formation due to their susceptibility to erosion over time
- **Tharsis Bulge** — a vast volcanic plateau on Mars approximately 5,000 km across and 10 km high, home to the largest volcanoes in the solar system and responsible for significant deformation of the Martian crust
- **Olympus Mons** — the largest volcano in the solar system, standing approximately 21.9 km above the Martian datum and spanning 600 km in diameter, located on the Tharsis Bulge
- **Chryse Planitia** — a large plain in the northern lowlands of Mars at approximately 28°N, 320°E, notable as the landing site of Viking 1 and as the terminus of several ancient outflow channels suggesting large-scale ancient flooding

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Landing Site Selection

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